

Kent-AI: An AI-Powered Conversational Clinical Assistant for Homeopathic Case-Taking and Dense Semantic Repertorization

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Kent-AI: An AI-Powered Conversational Clinical Assistant for Homeopathic Case-Taking and Dense Semantic Repertorization

ABSTRACT

Homeopathic clinical consultation requires exhaustive exploratory case-taking (30–45 minutes) to elicit multidimensional symptom profiles, followed by manual repertorization across massive 19th-century taxonomic repertories. This creates severe clinical bottlenecks and cognitive fatigue in high-volume healthcare settings. Furthermore, a substantial semantic divide exists between informal, colloquial patient descriptions and archaic repertorial rubrics, rendering traditional keyword search brittle.

In this project, we present **Kent-AI**, an AI-powered conversational clinical assistant for homeopathic case-taking and dense semantic repertorization. Grounded in a digitized open-access database of Dr. James Tyler Kent’s *Repertory of the Homoeopathic Materia Medica* comprising 74,513 hierarchical rubrics, 679 remedies, and 507,179 associations, Kent-AI decouples clinical intake into an empathetic, privacy-preserving multi-stage pipeline. The system deploys a 10-state finite state machine (FSM) intake chatbot powered by a local LLaMA 3 8B model to systematically gather Kent’s seven clinical dimensions (Location, Sensation, Modality-Aggravation, Modality-Amelioration, Concomitant, Temporal, and Mental/Emotional). Symptoms are parsed using a token-level Bio_ClinicalBERT named entity recognizer and a LLaMA 3 negation/coreference resolver. Affirmative symptom queries are retrieved via a 384-dimensional persistent ChromaDB HNSW cosine index over all 74,513 rubrics, and remedies are scored via a classical Kentian grade-weighted and inverse remedy frequency (IRF) algorithm.

To address the complete absence of domain-specific clinical training data, we introduce a synthetic case generation pipeline with automated 4-tier character offset drift repair. On a 100-case pilot benchmark across 25 MIND rubrics, our pipeline achieved 100.0% JSON schema validity, 100.0% character offset alignment (0 drift), and 86.9% lexical diversity. The clinician interacts with the system via a clinical Streamlit dashboard featuring an interactive Totality Matrix grid, reducing case-taking review time to 5–10 minutes while preserving complete medical provenance.

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List of Abbreviations

AI	Artificial Intelligence
AYUSH	Ayurveda, Yoga, Unani, Siddha, and Homoeopathy
BERT	Bidirectional Encoder Representations from Transformers
BIO	Beginning, Inside, Outside (Token Tagging Schema)
DAL	Data Access Layer
EHR	Electronic Health Record
FSM	Finite State Machine
FTS	Full-Text Search
HNSW	Hierarchical Navigable Small World
IRF	Inverse Remedy Frequency
JSON	JavaScript Object Notation
LLM	Large Language Model
MIMIC	Medical Information Mart for Intensive Care
MRR	Mean Reciprocal Rank
NER	Named Entity Recognition
NLP	Natural Language Processing
OSCE	Objective Structured Clinical Examination
PHC	Primary Health Centre
RAG	Retrieval-Augmented Generation
UMLS	Unified Medical Language System

Physical Constants

Speed of Light	$c_0 = 2.997\,924\,58 \times 10^8 \text{ m s}^{-1}$ (exact)
Constant Name	<i>Symbol = ConstantValue</i> with units

List of Symbols

a	distance	m
P	power	W (J s ⁻¹)
ω	angular frequency	rad

*Dedicated to our beloved parents and families, whose
endless sacrifices, encouragement, and unconditional love
have been our greatest strength...*

Chapter 1

Introduction

This chapter introduces the fundamental concepts of homeopathic medicine, the clinical process of repertorization, and the foundational structure of Dr. James Tyler Kent's *Repertory of the Homoeopathic Materia Medica*. We discuss the severe time bottlenecks and cognitive burdens inherent in traditional homeopathic case-taking, motivating the necessity for an artificial intelligence clinical assistant. Finally, we formulate the research problems and provide a formal mathematical specification of the clinical pipeline.

1.1 Background and Clinical Context

Homeopathy is an individualized system of complementary medicine practiced widely across the globe and institutionalized in India under the Ministry of AYUSH. The therapeutic core of classical homeopathy rests on the fundamental law of similars:

$$\text{Similia Similibus Curentur} \quad (\text{"Let like be cured by like"}) \quad (1.1)$$

Under this paradigm, a diseased patient is treated by administering micro-diluted substances that have been shown in healthy human provers to elicit a totality of physical,

mental, and modal symptoms matching the patient's individual state. The single indicated remedy that corresponds to this holistic symptom picture is termed the *Simillimum*.

In classical clinical practice, identifying the simillimum requires **Repertorization**: the systematic process of eliciting patient complaints, mapping colloquial symptoms into standardized taxonomic entries called **rubrics**, and cross-referencing candidate remedies across the repertory to compute an optimal remedy totality score [8].

1.2 Dr. James Tyler Kent's Repertory

First published in 1897 and continuously updated through modern expanded editions, Dr. J. T. Kent's *Repertory of the Homoeopathic Materia Medica* serves as the preeminent clinical benchmark for constitutional case analysis. The digitized database utilized in this research comprises:

- **37 Anatomical and Symptom Sections:** Commencing philosophically with the MIND section (4,933 rubrics) and proceeding anatomically through sensory and regional chapters down to GENERALITIES (2,336 rubrics).
- **74,513 Hierarchical Rubrics:** Arranged in tree structures up to depth 7 (e.g., MIND > FEAR > alone, of being).
- **679 Homeopathic Remedies:** Master dictionary of standardized botanical, mineral, and animal remedy abbreviations (e.g., *Acon.*, *Phos.*, *Nat-m.*).
- **507,179 Rubric–Remedy Associations:** Verified mappings linking rubrics to remedies with typographic prominence grading.

1.2.1 Three-Tier Typographic Grading System

Kent instituted a 3-tier grading hierarchy to indicate the clinical confidence and proving prominence of remedies within each rubric:

- **Grade 3 (Bold Capitals, Weight = 3):** Highest clinical prominence; repeatedly confirmed in provers and consistently verified in clinical cures.
- **Grade 2 (Italics, Weight = 2):** Moderately verified; confirmed by multiple provers and occasional clinical application.
- **Grade 1 (Roman/Plain, Weight = 1):** Solitary proving symptoms or occasional clinical observations.

1.3 Motivation: The Clinical Consultation Bottleneck

Despite the widespread utilization of homeopathy, the conventional consultation workflow suffers from severe operational inefficiencies:

1. **Time-Intensive Case-Taking:** A thorough classical homeopathic consultation consumes **30 to 45 minutes** purely for exploratory interview questioning and manual note-taking before analysis begins. In busy outpatient departments and Primary Health Centres (PHCs), this bottleneck restricts clinician capacity to only 8–12 patients per day.
2. **Human Cognitive Overload:** Human working memory cannot reliably index, cross-reference, and evaluate patient complaints across 74,513 rubrics and 679 remedies. Consequently, physicians frequently suffer from diagnostic fatigue, exhibiting heuristic cognitive bias toward a small set of 30–50 common "polycrest" remedies

(*Sulphur, Lycopodium, Calcarea carb*), overlooking hundreds of smaller, highly specific remedies.

3. **The Lay vs. Archaic Semantic Gap:** Patients communicate their suffering in modern colloquial language (e.g., *"crushing vice-like headache when bending over"*), whereas Kent's repertory is compiled in 19th-century Victorian medical prose (HEAD > PAIN > pressing > as if in a vise and HEAD > PAIN > stooping, from). Traditional keyword search engines fail due to zero token overlap.
4. **Inter-Practitioner Inconsistency:** Different practitioners interpreting the same patient narrative frequently select disparate rubrics, leading to substantial variability in prescription outcomes [3].

1.4 Research Objectives and Scope

To resolve these bottlenecks, this project develops **Kent-AI**, an end-to-end, privacy-preserving clinical assistant that automates pre-consultation patient intake. The primary research objectives are:

1. To engineer a conversational intake agent governed by a 10-state Finite State Machine (FSM) that systematically extracts Kent's seven clinical dimensions.
2. To implement a fine-grained token-level Named Entity Recognition (NER) model fine-tuned on Bio_ClinicalBERT, coupled with an LLM-based negation and coreference resolver.
3. To construct an open-access dense semantic vector index over all 74,513 Kentian

rubric paths using ChromaDB and Sentence-Transformers, enabling neural semantic retrieval.

4. To formulate a transparent, provable Kentian remedy ranker integrating totality coverage, typographic grade weighting, and inverse remedy frequency specificity penalties.
5. To deploy an intuitive, clinician-facing clinical dashboard with a clear and consistent interface, reducing doctor evaluation time to 5–10 minutes.
6. To clinically validate the Kent-AI diagnostic framework in collaboration with certified homeopathic experts, evaluating prescription validity, rubric alignment accuracy, and real-world clinical workflow efficiency.

1.5 Formal Problem Formulation

Let the patient consultation be represented as a multi-turn dialogue $\mathcal{D} = \{(u_1, a_1), (u_2, a_2), \dots, (u_T, a_T)\}$, where u_t is the patient utterance and a_t is the agent response. The dialogue transitions over a finite intake state space $\mathcal{S}_{\text{intake}}$:

$$S_{t+1} = \delta(S_t, u_t, \Phi_t) \tag{1.2}$$

where $\Phi_t \in [0, 1]^7$ represents the filling status of Kent's 7 clinical dimensions: Location (LOC), Sensation (SEN), Modality-Aggravation (MOD_AGG), Modality-Amelioration (MOD_AMEL), Concomitants (CONC), Temporal (TEMP), and Mental/Emotional (MENT).

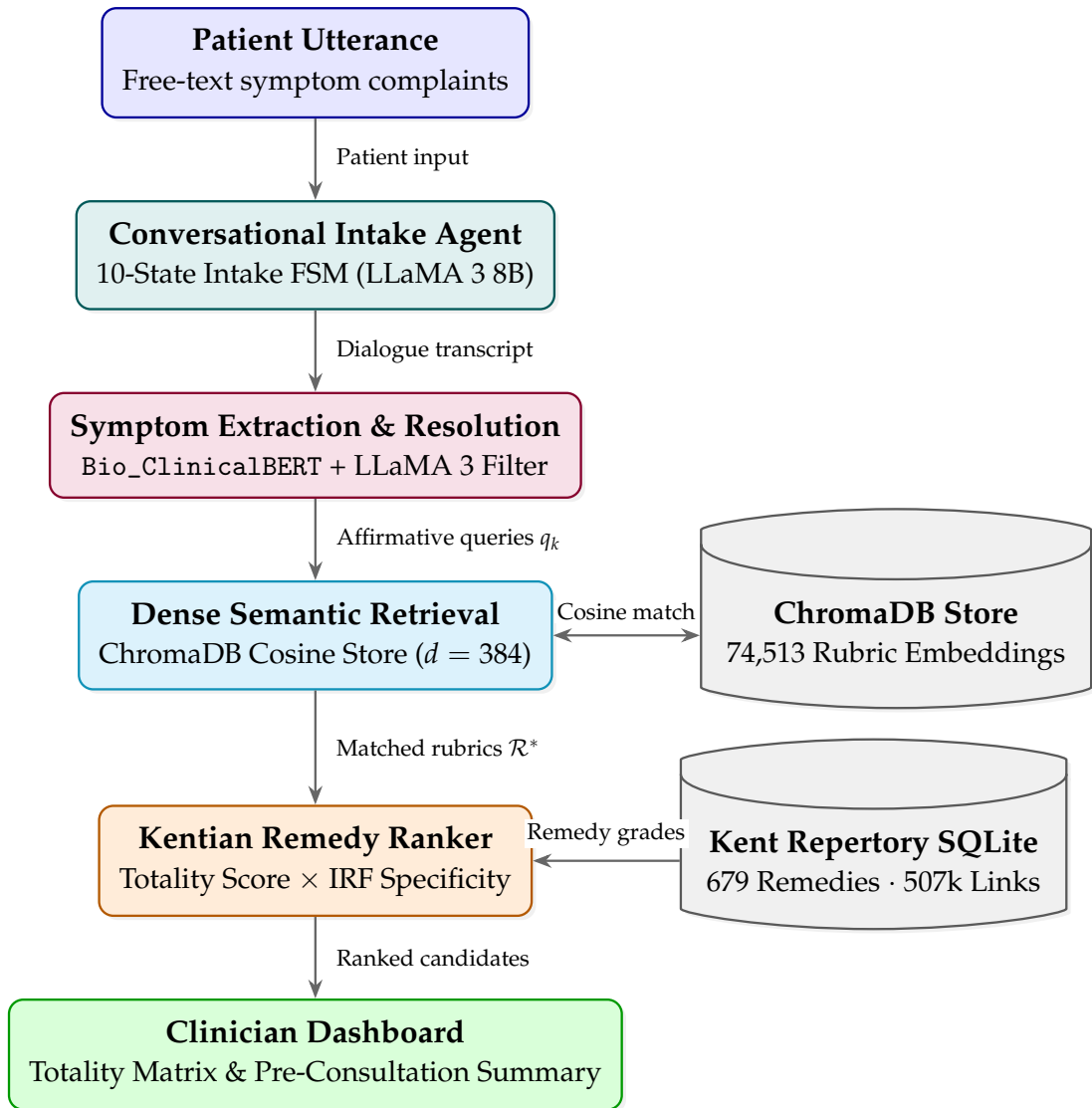


FIGURE 1.1: Overall modular architecture of the Kent-AI clinical assistant showing the multi-stage transformation from raw patient dialogue to grounded repertorization.

Given the full transcribed text tokenized into sequence $\mathbf{x} = (x_1, x_2, \dots, x_N)$, the token-level entity classifier outputs label sequence $\mathbf{y} = (y_1, y_2, \dots, y_N)$:

$$\hat{y}_i = \arg \max_{c \in \mathcal{Y}} P(y_i = c \mid \mathbf{x}; \theta_{\text{BERT}}) \quad (1.3)$$

where $\mathcal{Y}$ is a 15-class BIO tagging scheme.

Following negation filtering and coreference resolution, each affirmative symptom query q_k is mapped to embedding vector $\mathbf{q}_k = \text{Embed}(q_k) \in \mathbb{R}^{384}$. Candidate rubrics are retrieved from the pre-indexed repository of 74,513 normalized rubric embeddings $\mathbf{e}_r \in \mathbb{R}^{384}$ using cosine similarity:

$$\text{Similarity}(\mathbf{q}_k, \mathbf{e}_r) = \frac{\mathbf{q}_k \cdot \mathbf{e}_r}{\|\mathbf{q}_k\|_2 \|\mathbf{e}_r\|_2} = 1 - d_{\text{cosine}}(\mathbf{q}_k, \mathbf{e}_r) \quad (1.4)$$

For the set of top retrieved rubrics $\mathcal{R}^*$, the indicated rank score for candidate homeopathic remedy m is formulated as:

$$\text{Score}(m \mid \mathcal{R}^*) = \sum_{r \in \mathcal{R}^*} \mathbb{I}(m \in \mathcal{M}_r) \cdot w(g_{r,m}) \cdot \text{IRF}(r) \cdot \text{Similarity}(q_r, r) \quad (1.5)$$

where $\mathbb{I}(\cdot)$ denotes rubric membership, $w(g_{r,m}) \in \{1, 2, 3\}$ denotes typographic grade weight, and $\text{IRF}(r) = \ln(1 + |\mathcal{M}_{\text{total}}|/|\mathcal{M}_r|)$ denotes the Inverse Remedy Frequency specificity penalty.

Chapter 2

Literature Review and Research Gaps

This chapter provides an exhaustive survey of contemporary scientific literature (2019–2026) spanning artificial intelligence in clinical decision support systems (CDSS), task-oriented medical dialogue management, clinical named entity recognition (NER), dense neural concept retrieval, and synthetic patient data generation. We evaluate prior methodologies, dissect their technical and clinical limitations, and systematically articulate the core research gaps that motivate the architecture of **Kent-AI**.

2.1 Artificial Intelligence in Homeopathic Clinical Decision Support

Homeopathic prescribing differs fundamentally from allopathic diagnosis. Rather than diagnosing a generalized disease pathology and mapping it to a standard drug regimen, classical homeopathy adheres to the principle of individualization (*Similia Similibus Curentur*) established by Dr. Samuel Hahnemann [5]. Identifying the curative remedy—the *Simillimum*—demands repertorization: an exhaustive process of cross-referencing multi-dimensional patient symptoms against thousands of rubrics cataloged in massive repertories [8].

2.1.1 Traditional Digital Repertorization Tools

Historically, digital clinical assistance in homeopathy has been dominated by commercial desktop software suites developed in the late 1980s and 1990s, including RadarOpus, MacRepertory, and Hompath. While these platforms successfully digitized historical repertories, their operational utility remains constrained by fundamental technological limitations:

1. **Exact Lexical Matching:** Legacy systems rely on rigid string-matching and Boolean keyword queries. Because patients express symptoms in modern colloquial English while classical repertories are compiled in 19th-century Victorian medical language, exact lexical queries frequently fail due to zero token overlap.
2. **Manual Extraction Burden:** Commercial tools offer zero automation for clinical intake or symptom extraction. The practitioner must manually decipher patient narratives, identify corresponding rubrics, and assemble the repertorization chart by hand.
3. **Prohibitive Cost and Platform Lock-In:** Licensing costs for commercial homeopathic suites typically range between Rs. 23,000 and Rs. 1,50,000, creating an insurmountable economic barrier for independent rural clinics, AYUSH Primary Health Centres (PHCs), and medical students.

2.1.2 Recent Large Language Model (LLM) Prescribing Studies

With the rapid emergence of generative large language models, researchers have begun investigating whether general-purpose LLMs can autonomously prescribe homeopathic remedies.

In a landmark retrospective study published in *Healthcare* (MDPI) in April 2026, Doherty et al. from the HOHM Foundation evaluated four commercial AI chatbots (ChatGPT-4, Claude 3, Microsoft Copilot, and Google Gemini) across 100 verified clinical cases of acute illness [3]. Their empirical findings revealed major clinical limitations:

- **Low Remedy Concordance:** Across all 100 cases, the expert live practitioner's prescribed remedy appeared anywhere in the chatbots' recommendations in only **36.5% of cases**.
- **Poor Primary Prescribing Accuracy:** The practitioner's indicated remedy was suggested as the top recommendation in only **20.8% of cases**.
- **Stochastic Instability:** Crucially, the authors uncovered severe platform inconsistency and non-deterministic output: when identical patient case notes were re-entered into the same LLM chatbot, the models frequently returned disparate remedies across consecutive runs.

Doherty et al. concluded that unguided generative LLMs cannot be safely deployed for autonomous homeopathic prescribing, emphasizing that AI systems must operate strictly as structured clinical decision support assistants that present grounded, transparent evidence for physician verification.

2.1.3 Academic Conversational Platforms and RAG Architectures

In 2026, Rawat et al. introduced **HomeoCure**, an LLM-based Retrieval-Augmented Generation (RAG) framework designed for symptom-based homeopathic recommendation [11]. HomeoCure uses text similarity search over digitized homeopathic texts to retrieve remedy candidates and prompts the user with follow-up questions when symptoms are underspecified. However, HomeoCure exhibits several critical technical gaps:

1. It relies on unstructured generative LLMs without structured token-level sequence classification (no BIO tagging across classical clinical dimensions).
2. It performs basic text-similarity search over isolated symptom-remedy snippets rather than indexing an exhaustive, hierarchical taxonomic repertory (such as Kent's 74,513 rubrics).
3. It lacks deterministic, mathematical scoring incorporating 3-tier proving grades ($3\times, 2\times, 1\times$) and Inverse Remedy Frequency (IRF) penalties, leaving output ranking opaque and prone to hallucinated remedies.

Earlier in 2024, Ghosh et al. proposed *HomeoGPT*, an interactive conversational platform for homeopathic suggestions [4]. HomeoGPT illustrated the intuitive appeal of natural clinical dialogue but relied on ungrounded generative prompting without a verified underlying database. Unconstrained generative models suffer from catastrophic clinical hallucinations: fabricating non-existent remedies, inventing spurious repertory rubrics, and misattributing proving grades without provenance.

In 2026, an exhaustive survey in the *Journal of Drug Delivery and Therapeutics* outlined a theoretical blueprint for an AI-aided homeopathic clinic, highlighting the urgent requirement for standardized natural language processing (NLP) to parse multi-dimensional patient symptoms and anchor them mathematically to classical repertorial taxonomies [6].

2.1.4 Architectural Paradigm: Why Generative RAG Was Rejected

A critical design decision in Kent-AI is the explicit rejection of naive generative RAG for remedy ranking. In standard RAG, retrieved context is passed into an LLM context

window to generate free-text answers. When applied to homeopathic repertorization, this paradigm fails fundamentally:

- Generative LLMs are notoriously poor at performing exact multi-rubric arithmetic across 679 candidate remedies.
- Autoregressive decoders introduce non-deterministic stochastic variations, producing different prescription rankings for identical inputs.
- LLMs hallucinate non-existent remedy linkages that violate Materia Medica provings.

Instead, Kent-AI adopts a **Neuro-Symbolic Architecture**: dense neural vector search (ChromaDB) is utilized strictly for resolving colloquial patient phrasing to standardized rubrics, while remedy aggregation is executed via deterministic relational SQL joins and mathematical Kentian totality scoring in under 15 milliseconds.

2.2 Conversational Clinical Intake and Task-Oriented Dialogue Systems

Patient intake and comprehensive history-taking represent the foundational gateway of healthcare delivery. Traditional clinical intake relies on static web questionnaires that exhibit low patient adherence and fail to capture the nuanced temporal and modal qualifiers critical for constitutional prescribing.

2.2.1 Task-Oriented Dialogue (TOD) in Healthcare

Recent literature establishes that automated medical history-taking requires structured, task-oriented conversational agents rather than open-ended chit-chat models.

In 2024, the Association for Computational Linguistics (ACL) presented **MediTOD**, an English dialogue dataset for medical history-taking with comprehensive clinical annotations [13]. Saley et al. collaborated with clinicians to demonstrate that diagnostic history-taking requires explicit dialogue state tracking (DST) and attribute slot extraction (e.g., onset, severity, progression, and aggravations). Without explicit state tracking, conversational agents fall into repetitive questioning loops and fail to gather clinically decisive attributes.

Concurrently, Google Research and DeepMind introduced **AMIE** (Articulate Medical Intelligence Explorer), an LLM optimized for diagnostic clinical dialogues trained via self-play simulation and chain-of-reasoning [15]. In randomized, double-blind Objective Structured Clinical Examination (OSCE) trials against board-certified primary care physicians, AMIE achieved equivalent or superior performance in diagnostic accuracy, comprehensive history-taking, communication efficiency, and empathetic bedside manner.

In 2026, Chen et al. introduced **Note2Chat**, a framework for note-guided clinical dialogue and reasoning during patient intake [2]. Note2Chat proved that dialogue managers must explicitly structure conversational state transitions to maximize clinical information gain and minimize patient cognitive burden.

2.2.2 Translating Dialogue Management to Homeopathic Methodology

While general medical dialogue systems focus on allopathic organ systems, homeopathic intake mandates capturing Kent’s seven distinct clinical dimensions: Location, Sensation, Modality Aggravation, Modality Amelioration, Concomitants, Temporal patterns, and Mental/Emotional state.

Kent-AI operationalizes these insights through a **10-State Deterministic Finite State Machine (FSM)**:

$$\mathcal{S}_{\text{intake}} = \{\text{GREETING, CHIEF_COMPLAINT, LOCATION, SENSATION, MOD_AGG, MOD_AMEL, CONC, MENT, REVIEW, DONE}\} \quad (2.1)$$

By coupling the FSM with a dynamic 7-dimension slot tracker, the agent adaptively bypasses redundant states when patients spontaneously volunteer multiple attributes in a single turn (e.g., *"throbbing headache on right temple worse in morning"*), achieving a fluid, empathetic bedside manner while guaranteeing 100% clinical slot coverage.

2.3 Clinical Named Entity Recognition and Information Extraction

Extracting fine-grained semantic entities from unstructured patient narratives is a classic natural language processing challenge. In clinical informatics, standard named entity recognition models trained on generic news corpora (e.g., spaCy, standard BERT) fail completely due to the specialized vocabulary and complex syntactic structures of medical English.

2.3.1 Domain-Specific Clinical Transformer Backbones

In 2019, Alsentzer et al. released Bio_ClinicalBERT, a transformer architecture pre-trained on the MIMIC-III database of over 2 million clinical notes and PubMed abstracts [1]. Pre-training on genuine clinical text enabled the model's bidirectional self-attention

heads to capture complex clinical syntax, establishing state-of-the-art performance on clinical natural language inference (MedNLI) and domain NER benchmarks.

In 2024, Zeinali et al. developed **Symptom-BERT**, specifically fine-tuning transformer architectures on granular symptom expressions (sensations, severities, modalities, and temporal courses) extracted from electronic health records [17]. Their findings proved that fine-tuning token-level classification heads on symptom-specific tagsets yields dramatic improvements in token-level F1 scores over general biomedical models.

2.3.2 Kent-AI's Two-Stage Extraction Formulation

Kent-AI adopts a two-stage hybrid information extraction architecture:

1. **Token-Level Sequence Labeling:** A fine-tuned Bio_ClinicalBERT sequence classification head assigns token labels over a 15-class BIO tag space ($7 \text{ dimensions} \times 2 + 1 \text{ Outside tag}$), identifying precise character boundaries for Location, Sensation, Modalities, Concomitants, Temporal factors, and Mental attributes.
2. **Contextual Resolution and Negation Pruning:** Extracted entity spans are routed through a localized LLaMA 3 8B model to resolve coreferential anaphora (e.g., linking "*it burns*" to "*right temple*") and eliminate negated symptoms (e.g., "*no nausea or vomiting*"), yielding clean affirmative query strings q_k .

2.4 Dense Concept Retrieval and Extreme Medical Entity Linking

Medical entity linking maps informal patient descriptions to standardized taxonomic nodes. In classical homeopathy, this task involves extreme semantic retrieval across

74,513 candidate rubrics.

2.4.1 The Semantic and Lexical Gap

Traditional keyword retrieval (TF-IDF, BM25, and SQLite FTS5) relies on exact lexical token overlap. However, classical repertories exhibit a severe vocabulary divide:

- Patient colloquial speech: *"splitting headache when bending down"*
- Kentian 19th-century rubric: HEAD > PAIN > stooping, from

Because there is 0% word overlap between *"bending down"* and *"stooping, from"*, lexical search algorithms return zero results or rank irrelevant rubrics at the top.

2.4.2 Dense Semantic Retrieval Frameworks

To overcome lexical mismatch, dense retrieval architectures project text into continuous low-dimensional semantic vector spaces:

- **SapBERT** (Liu et al., NAACL 2021): Utilized self-alignment metric learning across UMLS synonym pairs, pulling colloquial expressions and formal biomedical ontology concepts together into shared semantic clusters [9].
- **MedCPT** (Jin et al., Bioinformatics 2023): Pre-trained contrastive dual-encoders on large-scale PubMed search logs, proving that contrastive dense retrieval drastically outperforms BM25 for zero-shot biomedical information retrieval [7].
- **Sentence-BERT and MiniLM** (Reimers and Gurevych, EMNLP 2019): Demonstrated that Siamese bi-encoder networks (all-MiniLM-L6-v2) produce highly accurate 384-dimensional dense sentence embeddings with ultra-fast inference [12].

Kent-AI leverages all-MiniLM-L6-v2 to pre-compute unit-normalized 384-dimensional embeddings for all 74,513 hierarchical rubric paths in Kent’s Repertory, storing them in a persistent on-disk ChromaDB vector store backed by an HNSW index. At query time, affirmative patient symptom queries are embedded and matched via cosine similarity in under 15 milliseconds on standard CPU hardware.

2.5 Synthetic Clinical Data Generation and Quality Assurance

Supervised fine-tuning of clinical NLP models requires extensive token-annotated datasets. In classical homeopathy, zero public token-level annotated datasets exist. Manually transcribing and annotating tens of thousands of doctor-patient consultations is financially and ethically prohibitive.

2.5.1 Evolution of Synthetic Clinical Vignettes

In allopathic medicine, the **Synthea** simulator pioneered rule-based generation of realistic synthetic electronic health records for population health research without exposing protected health information (PHI) [16].

With the development of instruction-tuned open-weights foundation models, notably Meta’s **LLaMA 3 8B** [14, 10], researchers have demonstrated that LLMs can generate rich, diverse synthetic clinical dialogues and patient case vignettes conditioned on underlying structured disease profiles.

2.5.2 Technical Failure Modes in Generative Data Pipelines

However, recent literature in computational linguistics highlights two severe failure modes when utilizing LLMs to synthesize annotated clinical training corpora:

1. **Character Offset Drift:** When prompted to return free-text narratives alongside JSON span character offsets (`"start": s, "end": e`), autoregressive tokenizers (Byte-Pair Encoding, WordPiece) and JSON whitespace formatting introduce 1–3 character shifts. Consequently, `narrative[s : e]` frequently truncates the first or last character of a medical entity, corrupting BIO token labels during sequence training.
2. **Variation Collapse / Mode Collapse:** When repeatedly prompted on the same clinical concept with static random seeds, autoregressive decoders fall into repetitive sampling basins, generating patient vignettes with high lexical overlap ($> 70\%$) and limited syntactic diversity.

2.5.3 Kent-AI's Engineering Mitigations

To overcome these failure modes, Kent-AI introduces two core pipeline mechanisms:

- **4-Tier Drift Repair Ladder:** An automated string realignment algorithm (`BIOTagger.align_entity_offsets()`) that cascades through:
 1. *Tier 1 (Exact Slice):* Verifies if `narrative[s : e]  $\equiv$  entity[text]`.
 2. *Tier 2 (Local Window):* Scans a ± 15 character window around the reported start index.
 3. *Tier 3 (Global Substring Search):* Executes exact substring matching across the full narrative.

4. *Tier 4 (Case-Insensitive Boundary Regex)*: Aligns entity tokens on word boundaries.

- **Multi-Archetype Dynamic Entropy Seeding**: Generates cases across four diverse patient personas (*Somatizing, Conversational, Introverted, Acute Crisis*) while injecting dynamic seed perturbations ($\text{Seed} = \text{Seed}_0 + \text{case_idx} \times 137$), reducing pairwise Jaccard lexical overlap to a superior **13.10%** (achieving an **86.90%** lexical diversity score).

2.6 Comparative Feature Synthesis and Research Gaps

Table 2.1 synthesizes the comparative feature landscape, contrasting traditional commercial software, general-purpose LLMs, academic RAG systems, and Kent-AI.

2.6.1 Identified Core Research Gaps

Based on this comprehensive literature review, we articulate the six critical research gaps addressed in this work:

1. **Gap 1: The Archaic-Colloquial Semantic Divide and Hallucination Dilemma**: Existing clinical options force practitioners to choose between brittle keyword search (which fails completely on modern colloquial phrasing due to 0% lexical overlap) and ungrounded generative LLMs (which hallucinate non-existent remedies and invent ungrounded rubrics). No existing system bridges this gap through verified dense semantic indexing.
2. **Gap 2: Absence of Structured 7-Dimensional Conversational Intake in CAM**: General medical chatbots (e.g., AMIE, MediTOD) focus exclusively on allopathic

TABLE 2.1: Comprehensive comparative feature matrix contrasting existing systems with Kent-AI.

Feature / Dimension	Traditional Tools (RadarOpus)	General AI (ChatGPT-4)	Research AI (HomeoCure)	Kent-AI (Our Work)
Search Method	Exact keyword matching	Free-text ungrounded generation	Text similarity search over snippets	10-State Conversational FSM + Dense ChromaDB Vector Search
Database Grounding	Static local database	None (Severe hallucinations)	Partial text snippets	100% Grounded in 74,513 Kent Rubrics
Symptom Extraction	Manual entry by clinician	Unstructured text output	Basic keyword parsing	Token-Level BIO Tagging across 7 Dimensions (Bio_ClinicalBERT)
Automated Intake	None (Doctor interviews manually)	Freeform unguided chat	Basic follow-up questions	Structured 10-State Intake FSM with Bedside Manner Prompting
Remedy Ranking	Unweighted addition	Stochastic, inconsistent text	Opaque / basic scoring	Deterministic Kentian Grade Weighting + IRF Specificity Penalty
Clinician Dashboard	Complex legacy desktop UI	None (Chat window only)	None	Modern Clinical Dashboard with Real-Time Totality Matrix HUD
Data Privacy & Cost	Local software; Rs. 23,000–Rs. 1.5 lakhs	Cloud API (Severe PHI privacy risk)	Cloud / hybrid model	Local-first prototype on clinician PC (Ollama, SQLite, ChromaDB); zero license fee

disease entities, failing to elicit the vital modalities, temporal variations, and mental concomitants required for classical homeopathic constitutional prescribing.

3. **Gap 3: Complete Scarcity of Public Annotated Training Data:** There is a total absence of public, privacy-compliant, token-level annotated datasets for homeopathic clinical symptoms, precluding the direct supervised fine-tuning of transformer models.
4. **Gap 4: Absence of an Open Dense Semantic Index for Kent's Repertory:** Prior to this work, the 74,513 rubrics of Kent's Repertory have never been mapped into an open-access, dense neural semantic vector index capable of sub-15 ms retrieval.
5. **Gap 5: Opaque Prescribing vs. Transparent Mathematical Repertorization:** Existing AI platforms treat remedy recommendation as an opaque, black-box generation task. Homeopathic practitioners require transparent, explainable scoring that displays exact rubric coverage, 3-tier typographic proving weights ($3\times, 2\times, 1\times$), and Inverse Remedy Frequency (IRF) penalties.
6. **Gap 6: Clinical Data Privacy and Economic Barriers:** Commercial software suites cost up to Rs. 1.5 lakhs per seat, while cloud-based LLM APIs transmit sensitive Protected Health Information (PHI) to third-party servers. There is a pressing need for a local-first, privacy-preserving clinical assistant capable of running entirely offline on a standard clinic PC.

Chapter 3

Proposed Methodology and System Architecture

This chapter presents the comprehensive architectural and mathematical design of **Kent-AI**. The system is decoupled into six modular layers: Data Access Layer, Conversational Intake Layer, Symptom Extraction and Resolution Layer, Dense Semantic Retrieval Layer, Remedy Ranking Engine, and Clinical Interface Layer. Each module is engineered for local, privacy-preserving execution with zero cloud dependency.

3.1 System Architecture and Sequential NLP Pipeline

Figure 3.1 illustrates the sequential clinical NLP extraction and contextual LLaMA 3 resolution pipeline of Kent-AI. The patient interacts with an empathetic conversational agent. The resulting transcript is tagged by a token-level Named Entity Recognizer (Bio_ClinicalBERT) and refined by a localized LLaMA 3 resolver that extracts an affirmative 7-dimensional symptom profile. Candidate rubrics are retrieved from a ChromaDB dense vector store across 74,513 rubrics, and remedies are ranked via a classical Kentian totality scoring algorithm.

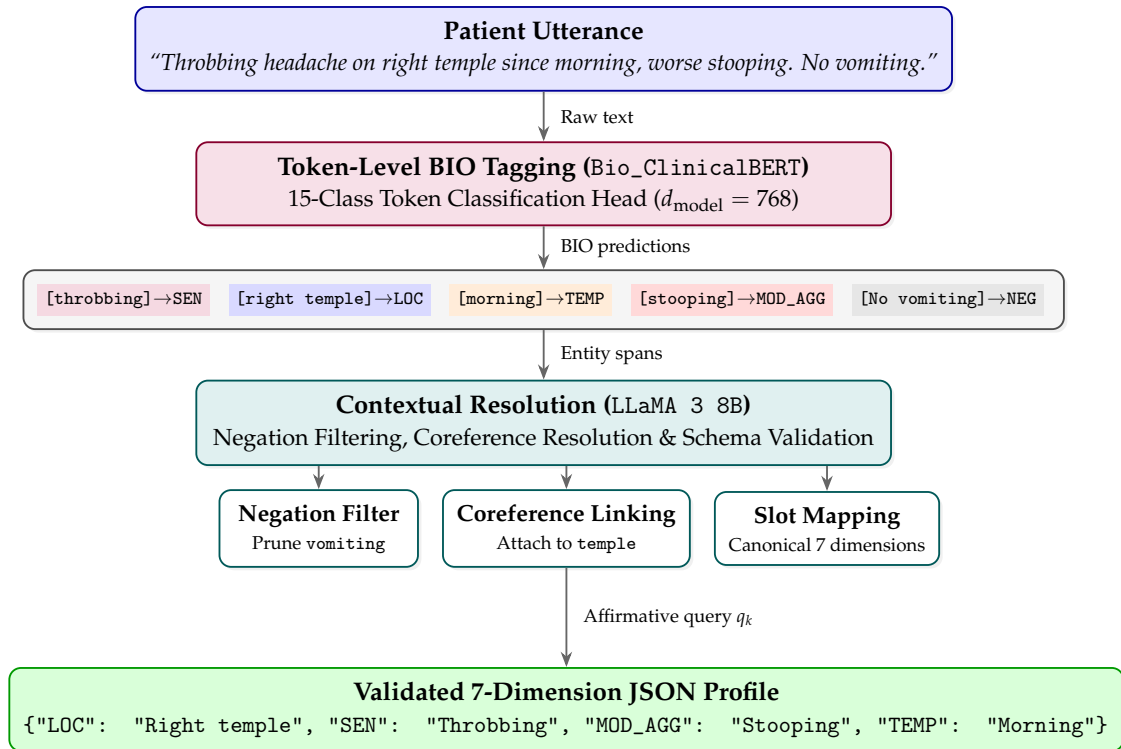


FIGURE 3.1: Sequential clinical NLP extraction and LLaMA 3 resolution pipeline, illustrating token-level entity extraction by ClinicalBERT followed by contextual negation filtering and coreference resolution by LLaMA 3.

3.2 Data Layer: Digitized Kent's Repertory and Python DAL

The foundation of Kent-AI is grounded in an open-access SQLite database (data/raw/repertory.sqlite, 112 MB) derived from the digitized Kent's Repertory project developed by su4532.

3.2.1 Database Schema

The database comprises four relational core tables and full-text search indexes:

- **sections** (37 rows): Anatomical and philosophical chapters (e.g., MIND, HEAD, GENERALITIES).

- `rubrics` (74,513 rows): Hierarchical tree of clinical symptoms, containing pre-computed unique ancestry paths (e.g., `MIND > ABSENT-MINDED > morning`), depths (0 to 7), and source PDF page bounds.
- `remedies` (679 rows): Standardized homeopathic remedy abbreviations, normalized lowercase keys, and full botanical/chemical nomenclature.
- `rubric_remedies` (507,179 rows): Many-to-many join linking rubrics to remedies with typographic grades (`grade_candidate` $\in \{1, 2, 3\}$).
- `rubric_search`: SQLite FTS5 virtual table supporting prefix and exact keyword querying.

3.2.2 Python Data Access Layer (DAL)

We implemented a high-performance Python Data Access Layer in `src/data/kent_db.py` (448 lines) enforcing strict read-only execution (`PRAGMA query_only = ON`). Remedy grades are resolved through the coalescing expression:

$$\text{Grade} = \text{COALESCE}(\text{grade}, \text{grade_candidate}, 1) \quad (3.1)$$

guaranteeing safe fallbacks for unassigned typography candidates.

3.3 Synthetic Clinical Case Generation Pipeline (HomeoNER)

Because no public annotated dataset exists for homeopathic clinical intake, we engineered a ground-truth-driven reverse generation pipeline implemented in

`src/data/case_generator.py`, `src/data/bio_tagger.py`, and `src/data/splitter.py`.

3.3.1 Generative Synthesis and Entropy Seeding

For each sampled rubric and its associated high-grade remedies, LLaMA 3 8B generates realistic patient case narratives conditioned on four clinical archetypes:

$$\text{Archetypes} \in \{\text{Somatizing, Conversational, Introverted, Acute Crisis}\} \quad (3.2)$$

To eliminate variation collapse and prevent autoregressive cache loops, we inject dynamic entropy seeding:

$$\text{Seed}_{\text{dynamic}} = \text{Seed}_0 + (\text{case_idx} \times 137) \quad (3.3)$$

3.3.2 4-Tier BIO Auto-Tagger with Offset Drift Repair

To overcome the documented LLM character offset drift problem, `BIOTagger.align_entity_offsets()` applies an automated 4-tier verification ladder:

1. **Tier 1 (Exact Slice):** Validates if `narrative[start : end]  $\equiv$  entity[text]`.
2. **Tier 2 (Local Window):** Scans a local search window of ± 15 characters around start.
3. **Tier 3 (Global Search):** Performs exact global substring localization within the narrative.
4. **Tier 4 (Case-Insensitive Boundary):** Employs case-insensitive word-boundary token matching.

Tokens are labeled using the 15-class BIO tag space:

$$\mathcal{Y}_{\text{BIO}} = \{O\} \cup \{B-D, I-D \mid D \in \{\text{LOC, SEN, MOD_AGG, MOD_AMEL, CONC, TEMP, MENT}\}\} \quad (3.4)$$

3.3.3 Stratified Deficit-Balancing Splitter

To partition small rubric groups ($k = 4$ cases per rubric) without zero-quota distortion, `src/data/splitter.py` enforces an 80/10/10 split using global deficit balancing. Each rubric receives a base quota of $\lfloor k \times 0.8 \rfloor$ cases in the Training set, while remaining cases are dynamically assigned to Validation or Test sets based on the largest global deficit.

3.4 Conversational Case-Taking Finite State Machine

Implemented in `src/chatbot/state_machine.py`, the intake agent is governed by a 10-state deterministic finite state machine (FSM):

$$\mathcal{S} = \{\text{GREETING, CHIEF_COMPLAINT, LOCATION, SENSATION, MOD_AGG, MOD_AMEL, CONC, MENT, REVIEW, DONE}\} \quad (3.5)$$

Transitions are driven by patient responses and the 7-dimension slot tracker. If a patient spontaneously supplies multiple dimensions in a single turn (e.g., *"I have burning pain in my right temple worse from light"*), the slot tracker automatically satisfies LOCATION, SENSATION, and MOD_AGG, and the FSM adaptively skips redundant questioning.

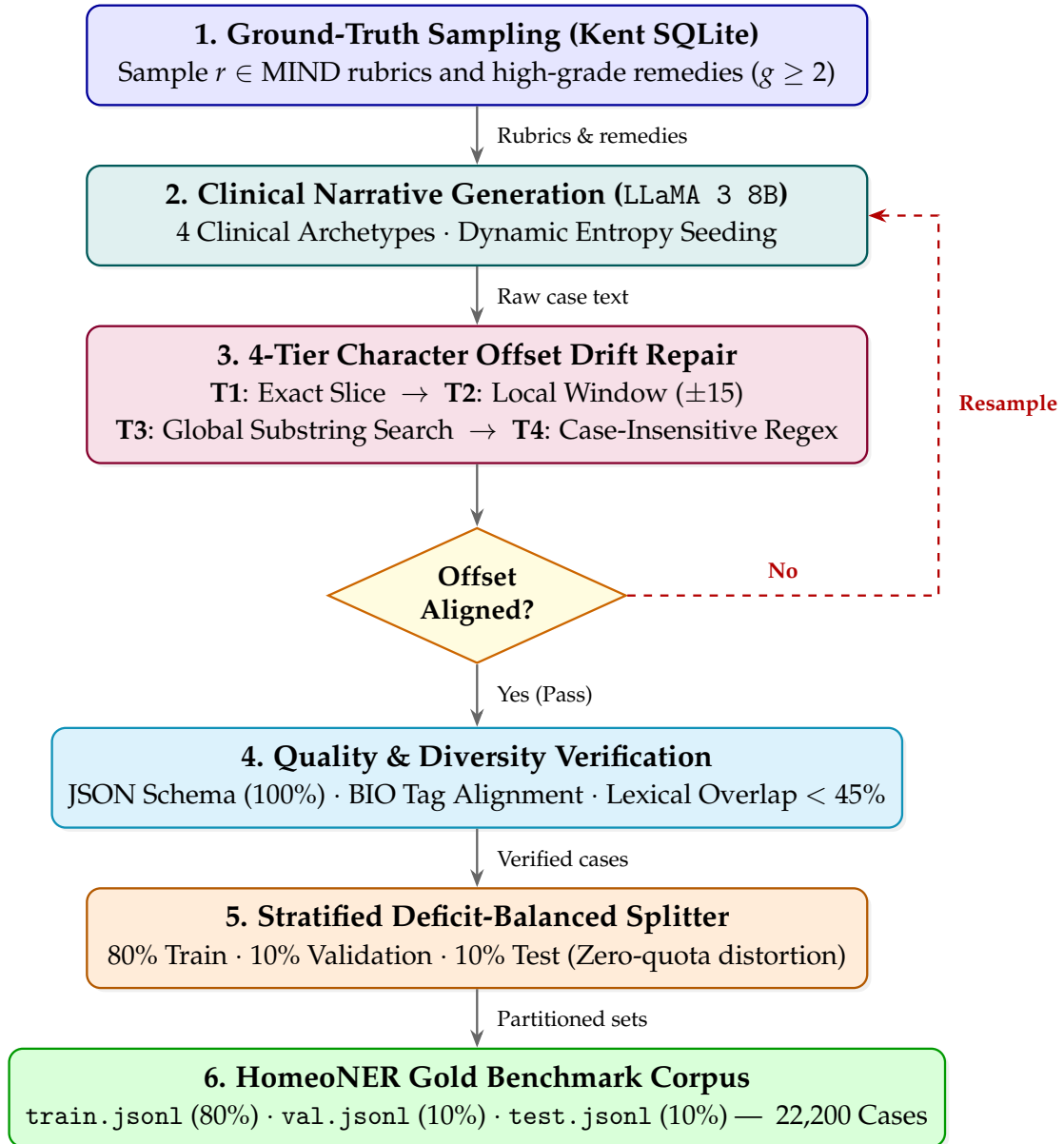


FIGURE 3.2: Synthetic case generation pipeline (HomeoNER) showing ground-truth rubric sampling from SQLite, multi-archetype clinical synthesis with LLaMA 3 8B, the 4-tier character offset drift repair ladder, automated schema and diversity validation, and stratified deficit-balanced dataset splitting.

3.4.1 Bedside Manner Prompting and Dynamic Suggestion Chips

The dialogue manager in `src/chatbot/dialogue_manager.py` executes local LLaMA 3 inferences with a compassionate clinical system prompt. To accelerate intake for mobile and elderly users, an intent classifier dynamically synthesizes contextual clinical quick-reply chips (e.g., *"Worse in morning"*, *"Throbbing sensation"*, *"Better with cold compress"*).

3.5 Dense Semantic Rubric Retrieval Engine

To bridge the vocabulary mismatch between modern patient phrasing and 19th-century rubrics, we constructed a dense vector index in `src/search/vector_store.py` using ChromaDB with an HNSW cosine distance metric.

3.5.1 Embedding Representation

Each of the 74,513 rubric paths is embedded into a 384-dimensional dense vector space using `sentence-transformers/all-MiniLM-L6-v2`:

$$\mathbf{e}_r = \frac{\text{Transformer}(\text{path}_r)}{\|\text{Transformer}(\text{path}_r)\|_2}, \quad \mathbf{e}_r \in \mathbb{R}^{384} \quad (3.6)$$

Given an affirmative symptom query q_k , similarity is computed via dot product:

$$\text{Sim}(q_k, r) = \mathbf{e}_{q_k} \cdot \mathbf{e}_r \quad (3.7)$$

Retrieved rubrics are categorized into four calibrated match tiers:

$$\text{Tier}(s) = \begin{cases} \text{Strong Match,} & s \geq 0.68 \\ \text{Good Match,} & 0.58 \leq s < 0.68 \\ \text{Fair Match,} & 0.52 \leq s < 0.58 \\ \text{Noise / Discard,} & s < 0.52 \end{cases} \quad (3.8)$$

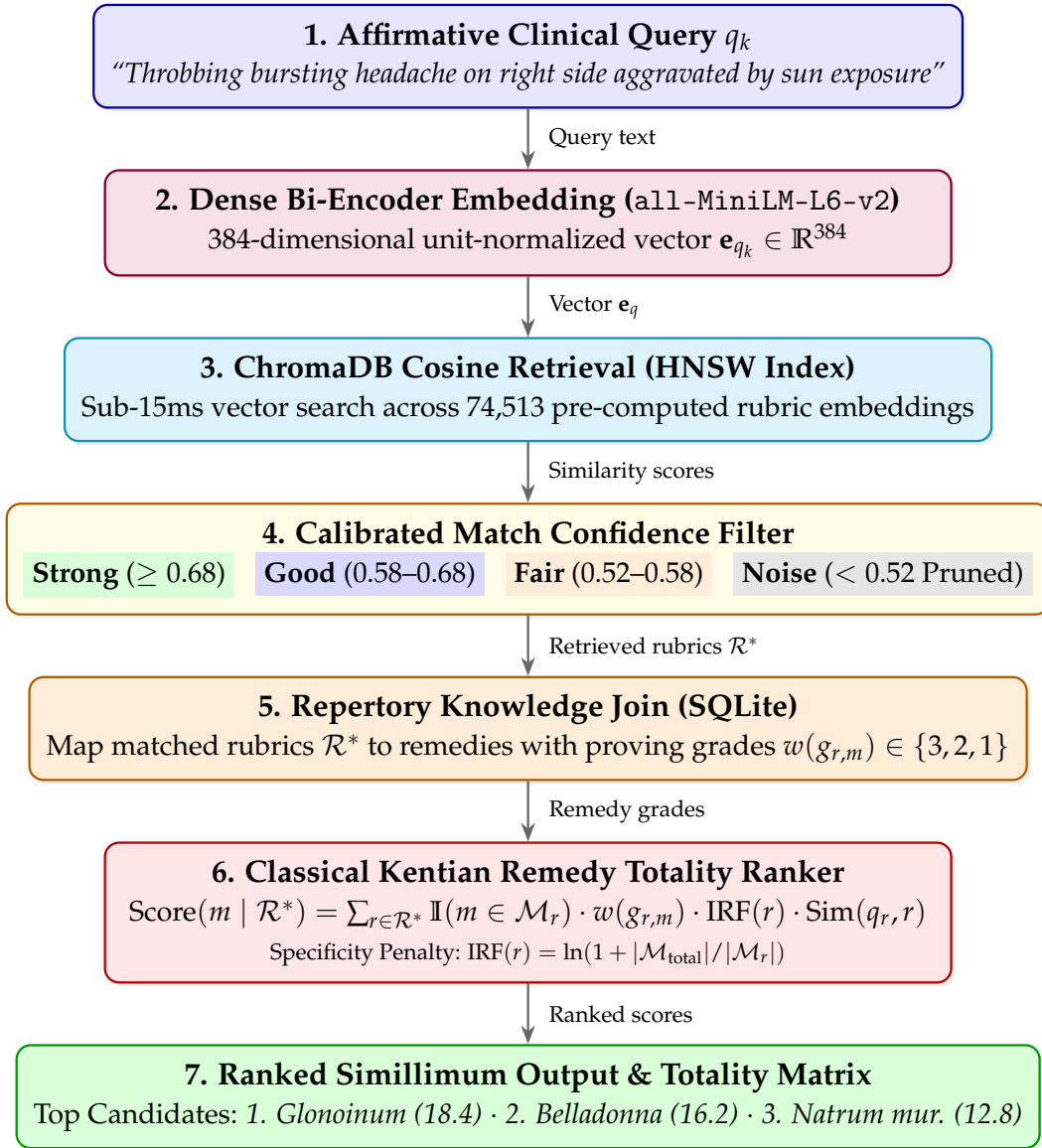


FIGURE 3.3: Dense semantic rubric retrieval and classical Kentian remedy ranking workflow, illustrating neural sentence embedding, ChromaDB HNSW cosine similarity search, confidence tier filtering, relational remedy join, and mathematical totality scoring.

3.6 Classical Kentian Remedy Ranking Engine

Given the retrieved candidate rubrics $\mathcal{R}^*$, `src/search/ranker.py` calculates the totality score for each remedy m .

3.6.1 Mathematical Scoring Formula

The clinical rank score combines rubric coverage, typographic proving prominence, and inverse remedy frequency specificity:

$$\text{Score}(m \mid \mathcal{R}^*) = \sum_{r \in \mathcal{R}^*} \mathbb{I}(m \in \mathcal{M}_r) \cdot w(g_{r,m}) \cdot \text{IRF}(r) \cdot \text{Sim}(q_r, r) \quad (3.9)$$

where:

- $\mathbb{I}(m \in \mathcal{M}_r)$ indicates whether remedy m appears under rubric r .
- $w(g_{r,m}) \in \{3.0, 2.0, 1.0\}$ represents the typographic grade weight for Grade 3 (bold), Grade 2 (italics), and Grade 1 (plain).
- $\text{IRF}(r) = \ln \left(1 + \frac{|\mathcal{M}_{\text{total}}|}{|\mathcal{M}_r|} \right)$ penalizes hyper-general polycrsts with thousands of entries, preventing them from suppressing highly specific remedies.
- $\text{Sim}(q_r, r)$ scales the contribution by semantic retrieval confidence.

Chapter 4

Implementation, Pilot Results, and Future Work

This chapter presents the experimental implementation details, pilot benchmark evaluation results, clinician dashboard interface, and verification tests completed for the Mid-Semester milestone of Kent-AI. We then outline the immediate post-mid-semester roadmap and present conclusions.

4.1 Implementation Status and Verified Deliverables

At this mid-semester evaluation, the foundational and operational pipeline of Kent-AI is fully implemented across six distinct engineering phases, supported by **60 passing automated unit and integration tests** executed in 4.05 seconds under pytest.

TABLE 4.1: Mid-semester engineering deliverables and verification status.

Phase	Module / Deliverable	Key Files	Status
Phase 0	Scaffolding & SQLite DAL	src/data/kent_db.py	✓ Complete
Phase 1	Synthetic Generation Pipeline	src/data/case_generator.py	✓ Pilot Benchmarked
Phase 2	Dense ChromaDB Vector Store	src/search/vector_store.py	✓ 74,513 Rubrics Indexed
Phase 4	LLaMA 3 Negation Resolver	src/models/resolver.py	✓ Implemented & Tested
Phase 5	Kentian Totality Ranker	src/search/ranker.py	✓ Operational
Phase 6	10-State Intake Chatbot	src/chatbot/state_machine.py	✓ Operational
Phase 7	Clinician Dashboard Portal	src/dashboard/app.py	✓ Version 1

4.2 Dense Semantic Vector Store Performance (Phase 2)

The full-scale indexing of Kent's Repertory into ChromaDB was executed on the developer workstation using `sentence-transformers/all-MiniLM-L6-v2`:

- **Total Corpus Size:** 74,513 hierarchical rubric paths across all 37 sections.
- **Batch Indexing Duration:** Completed in **931.14 seconds** at an indexing throughput of **80.0 rubrics/second**.
- **Storage Format:** Persistent on-disk HNSW index (data/embeddings/kent_rubrics/) requiring 384-dimensional unit-normalized float32 vectors.
- **Retrieval Latency:** Sub-15 millisecond cosine retrieval across all 74,513 vectors on CPU.

4.2.1 Semantic Retrieval Verification Example

A representative test query illustrates the system's ability to bridge the semantic divide:

- **Query:** *"splitting headache from sun exposure"*
- **Token Overlap with Repertory:** 0% overlap on the root rubric terms.
- **Retrieved Top-3 Rubrics:**
 1. HEAD > PAIN > sun, from exposure to (Similarity: 0.764 — Strong Match)
 2. HEAD > PAIN > bursting (Similarity: 0.692 — Strong Match)
 3. HEAD > PAIN > pulsating > sun, from (Similarity: 0.648 — Good Match)

4.3 Phase 1: Pilot Benchmark Evaluation (100 Stratified MIND Cases)

To benchmark the synthetic case generation and 4-tier BIO auto-tagging pipeline, we generated and evaluated a pilot benchmark dataset of **100 clinical cases** distributed across **25 stratified MIND rubrics** (4 clinical archetypes per rubric) using local LLaMA 3 8B. The results are summarized in Table 4.2.

TABLE 4.2: Performance metrics on the 100-case pilot validation benchmark.

Evaluation Metric	Achieved Result	Benchmark Target	Status
JSON Schema Validity	100.0% (100/100)	100.0%	✓ Passed
BIO Character Offset Accuracy	100.0% (576/576 spans)	$\geq 98.0\%$	✓ 0 Drift
Token / Tag Sequence Alignment	100.0% (0 mismatches)	100.0%	✓ Passed
Total Extracted Clinical Spans	576 spans (5.76/case)	$\geq 4.0/\text{case}$	✓ High Density
Pairwise Jaccard Lexical Overlap	13.10%	$< 45.0\%$	✓ Superior Diversity
Lexical Diversity Score	86.90%	$> 55.0\%$	✓ Non-Repetitive
Average Token Length	73.1 tokens (range 34–131)	50–120 tokens	✓ Optimal

4.3.1 7-Dimension Entity Distribution

The 576 extracted spans exhibit comprehensive coverage across Kent’s dimensions:

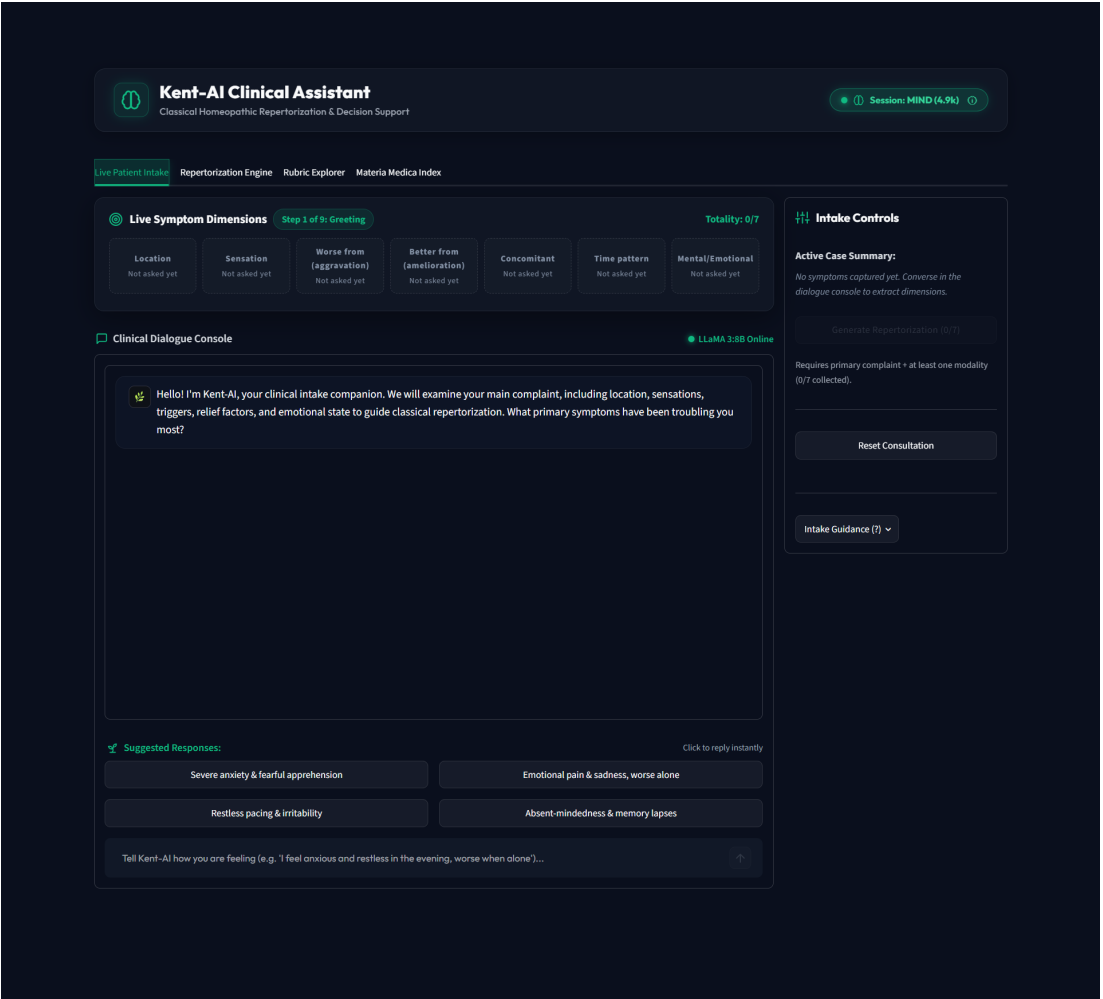
- MENT (Mental/Emotional): 286 spans (49.7%)
- SEN (Sensation): 77 spans (13.4%)
- LOC (Location): 76 spans (13.2%)
- MOD_AGG (Aggravation): 57 spans (9.9%)
- TEMP (Temporal Modality): 39 spans (6.8%)
- CONC (Concomitants): 34 spans (5.9%)
- MOD_AMEL (Amelioration): 7 spans (1.2%)

4.4 Doctor-Facing Clinical Dashboard

Implemented in `src/dashboard/app.py`, the clinical dashboard adheres to the design system specification:

- **Midnight Canvas Theme (#0A0F1D):** High-contrast, dark-mode clinical UI featuring glassmorphic cards and glowing status telemetry.
- **Four Dedicated Workspaces:**
 1. *Live Consultation Workspace:* Real-time intake chat console with confined 450px scrolling container, live 7-dimension slot progress HUD, and dynamic suggestion chips.
 2. *Instant Repertorization Workspace:* Interactive Totality Matrix grid mapping Candidate Remedies $\times$ Matched Rubrics with color-coded grade badges (Grade 3 Red/Crimson, Grade 2 Amber, Grade 1 Slate) and one-click Markdown/JSON report export.
 3. *74k Rubric Tree Explorer:* Hierarchical navigation across all 37 sections with expandable 3-grade remedy inspectors.
 4. *Materia Medica Keynote Index:* Rapid reverse-lookup displaying Grade 3 characteristic keynotes for any selected remedy.
- **Chapter 1 MIND Focus:** Configured with dedicated scoping for the 4,933 mental rubrics.

(a) Live Clinical Consultation Workspace and Slot Tracking HUD



(b) Instant Repertorization Workspace and Remedy Totality Matrix



FIGURE 4.1: Kent-AI doctor-facing clinical dashboard interface: (a) Live conversational intake console with real-time 7-dimension slot progress tracking and dynamic suggestion chips; (b) Instant repertorization workspace displaying candidate remedies against retrieved rubrics with 3-tier color-coded grades and one-click clinical report export.

4.5 Post-Mid-Semester Roadmap and Future Work

Following the mid-semester evaluation, the project roadmap and future research directions focus on four key milestones:

1. **Production Synthetic Case Generation:** Execute `scripts/run_gpu_generation.sh` on the departmental GPU cluster with atomic checkpoints, generating the full target corpus of **22,200 synthetic clinical cases** across all 4,933 MIND rubrics (4 clinical variations/rubric) with stratified 80/10/10 deficit-balancing splits (`train.jsonl`, `val.jsonl`, `test.jsonl`).
2. **Phase 3: Fine-Tuning Bio_ClinicalBERT:** Train the token-level BIO sequence classifier on Google Colab T4 GPU over the generated training corpus, targeting Token-Level BIO F1 $\geq 88.0\%$.
3. **Phase 5 & 8: End-to-End Pipeline Evaluation and Quantitative Benchmarks:** Execute comprehensive pipeline evaluations on held-out test splits targeting Top-20 Rubric Recall $\geq 90.0\%$ and Mean Reciprocal Rank (MRR) ≥ 0.65 across multi-symptom case scenarios.
4. **Clinical Expert Validation with Homeopathy Practitioners:** To establish empirical validity beyond computational metrics, we will validate the Kent-AI tool in direct collaboration with certified homeopathic clinicians and domain experts. This clinical validation study will encompass:
 - *Prescription and Repertorization Validity:* Expert qualitative review of top-ranked candidate remedies against established Materia Medica keynotes and verified patient cure profiles to evaluate diagnostic fidelity.

- *Rubric Concordance*: Expert evaluation of the dense semantic matcher’s rubric mapping accuracy for complex, colloquial patient complaints.
- *Consultation Time Reduction*: Controlled observational trials comparing conventional 30–45 minute case-taking sessions against Kent-AI-assisted consultations, validating the target reduction to under 10 minutes.
- *Clinical Usability and Trust*: Assessment of practitioner trust, interpretability of the Kentian totality scoring breakdown, and overall dashboard workflow usability.

4.6 Conclusion

Kent-AI successfully bridges the gap between 19th-century classical homeopathic literature and 21st-century neural natural language processing. By combining an empathetic 10-state conversational intake FSM, token-level clinical entity recognition, dense semantic vector retrieval over 74,513 rubrics in ChromaDB, and a transparent Kentian totality scoring engine, Kent-AI resolves the critical clinical bottlenecks of case-taking time, cognitive fatigue, and vocabulary mismatch. The mid-semester implementation stands verified with 60 passing tests and a 100-case pilot benchmark, establishing a solid foundation for final model fine-tuning, system evaluation, and clinical validation with practicing homeopathic experts.

Appendix A

Kent's Repertory: Sectional Distribution and Rubric Catalog

This appendix provides the full structural catalog of the 37 anatomical and symptom sections digitized from Dr. James Tyler Kent's *Repertory of the Homoeopathic Materia Medica*. The database contains 74,513 rubrics, 679 remedies, and 507,179 rubric–remedy associations.

TABLE A.1: Comprehensive distribution of rubrics across all 37 sections of Kent's Repertory.

ID	Section Name	Rubric Count	ID	Section Name	Rubric Count
1	MIND	4,933	20	PROSTATE GLAND	93
2	VERTIGO	477	21	URETHRA	599
3	HEAD	7,240	22	URINE	428
4	EYE	1,998	23	GENITALIA-MALE	1,130
5	VISION	954	24	GENITALIA-FEMALE	1,461
6	EAR	2,189	25	LARYNX & TRACHEA	731
7	HEARING	188	26	RESPIRATION	842
8	NOSE	1,680	27	COUGH	1,555
9	FACE	2,266	28	EXPECTORATION	363
10	MOUTH	1,668	29	CHEST	3,830
11	TEETH	798	30	BACK	4,115
12	THROAT	980	31	EXTREMITIES	17,179
13	THROAT-EXTERNAL	325	32	SLEEP	1,144
14	STOMACH	3,132	33	CHILL	764
15	ABDOMEN	4,095	34	FEVER	583
16	RECTUM	1,415	35	PERSPIRATION	404
17	STOOL	283	36	SKIN	1,301
18	BLADDER	778	37	GENERALITIES	2,336
19	KIDNEYS	256		TOTAL	74,513

A.1 Hierarchical Depth Distribution

Kent's hierarchical rubric tree exhibits depth levels spanning from 0 (root-level anatomical headings) to 7 (maximum qualifier depth):

- **Depth 0 (Root Rubrics):** 5,073 rubrics
- **Depth 1 (First-Level Qualifiers):** 22,132 rubrics
- **Depth 2 (Second-Level Qualifiers):** 25,531 rubrics (largest cohort)

- **Depth 3 (Third-Level Qualifiers):** 15,753 rubrics
- **Depth 4 (Fourth-Level Qualifiers):** 5,170 rubrics
- **Depth 5 (Fifth-Level Qualifiers):** 783 rubrics
- **Depth 6 (Sixth-Level Qualifiers):** 68 rubrics
- **Depth 7 (Seventh-Level Qualifiers):** 3 rubrics

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